

Conditional probability

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

1. Consider the chance experiment of rolling two dice, one red and one green. We can think about this experiment with our usual sample space from Example 1.2. Outcomes in the sample space are pairs (a, b) , where a is the number rolled on the red die and b is the number rolled on the green die.

Let A be the event “the sum of the numbers rolled is 8.” Let B be the event “the red die rolled a 3.”

- a. Write the sets of outcomes in the following events:

$$A = \{ \}$$

$$B = \{ \}$$

$$A \cap B = \{ \}$$

- b. Calculate $P(B \vee A)$ using the formula.

Your value for the conditional probability should make sense: because all the outcomes in the original sample space are equally likely, so are all the outcomes in A when we condition on the occurrence of A .

2. Julianne and Renaldo (thanks, Random Name Generator!) are playing a dice game. They each roll a die, and the person with the higher roll wins. If there is a tie, they roll again. Given that Julianne won, what is the probability that she rolled a 5?

To solve this problem systematically, use the Example 1.2 sample space, as in the first question above. Let the first roll be Julianne’s roll and the second roll be Renaldo’s roll. Let A be the event “Julianne won the game.” Let B be the event “Julianne rolled a 5.”

- a. Write the sets of outcomes in the following events:

$$A = \{ \}$$

$$B = \{1, 2, 3, 4, 5, 6\}$$

$$A \cap B = \{1, 2, 3, 4, 5, 6\}$$

b. Calculate $P(B \vee A)$ using the formula.

As in the first question, your value for the conditional probability should make sense: because all the outcomes in the original sample space are equally likely, so are all the outcomes in A when we condition on the occurrence of A .

3. Suppose we roll two dice. What is the probability that the sum of the numbers rolled is 7 given that neither die rolled a 6.

To answer this question, let A be the event “neither die rolled a 6,” and let B be the event “the sum of the numbers rolled is 7.” These events are subsets of the Example 1.2 sample space. You will calculate $P(B \vee A)$.

a. Write the elements of A . Which elements of A are also in B ?

b. When you condition on the occurrence of A , you reduce the sample space to A . All the outcomes in A are still equally likely. Given that fact, calculate $P(B \vee A)$.

4. I draw two cards from a shuffled deck. What is the probability that the second card is a heart given that the first card is a heart? First solve this problem systematically.

The sample space of equally likely outcomes is the set of ordered pairs (card 1, card 2) of distinct cards. Let A be the event “the first card is a heart” and B be the event “the second card is a heart.” You want to calculate $P(B \vee A)$.

a. How many outcomes are in the sample space?

b. Calculate the following probabilities in order, using the formula for the last probability:

$$P(A) = \frac{1}{2} \quad P(A \cap B) = \frac{1}{4} \quad P(B|A) = \frac{1}{2}$$